OPERATING MANUAL
FOR
PROGRAMMABLE MASTER CONTROLLER
AND
INSTALLATION MAINTENANCE MODULE

GEK-65442A

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SECTION 1
INTRODUCTION

PURPOSE OF MANUAL

This manual is a guide to the operation of General Electric's Programmable Master Controller (PMC) and Installation Maintenance Module (IMM). We have written this manual for both an inexperienced operator and one who is experienced but needs an occasional reminder of PMC or IMM operating functions.

We recommend that the inexperienced operator read the book from cover-to-cover to acquire an overview of the equipment operation. Thereafter, the book will be a useful reference manual and should be kept near the equipment.

AUDIENCE LEVEL

This publication is directed to experienced electronic technicians. Some experience in programming is preferable but not mandatory. The reader should be familiar with DIRECTO-MATIC® Plus equipment concepts and Control Block Language (CBL).

RELATED DOCUMENTS

Control Block Language Reference Manual GEK-65426
Field Support System Operator Manual GEK-65435
Control Signal Highway Maintenance Manual GEK-65411
Drive and Direct Control Equipment GEK-61680A
Hewlett-Packard 21MX E-Series Computer Installation and Service Manual.

SECTION 2
SYSTEM DESCRIPTION

FUNCTIONAL DESCRIPTION OF SYSTEM

A simple DIRECTO-MATIC® Plus system, built around the Programmable Master Controller (PMC) is shown in Figure 2-1. Such a system typically includes several operator stations which control a number of drives. The PMC normally performs the following functions:

- Coordinates the operation of drives through the use of operator commands.
- Displays to the operator the operations occurring in the system.
- Accepts inputs from system limit switches, pressure switches and other types of sensors and produces output commands to control motors and solenoids.

The communication between the PMC and other system equipment occurs via a Control Signal Highway (CSH). Each major station along the CSH is referred to as a slave station. A PMC can drive up to three separate Control Signal Highways. The interface control equipment located between the PMC and the CSH may operate as a master station (controlling the CSH) or as a slave (accepting information from an adjacent PMC).

The PMC exercises programmable control to coordinate system functions. In this case, the control system is programmed by use of "Control Block Language" (CBL). Control Block Language handles three types of control:

1. Sequencing or relay-type functions using the common relay language of the traditional programmable controller.
2. Analog or regulating-type functions using analog control blocks such as integrators or summers.
3. Certain special computing functions such as logging, simplified process setup, etc., through a procedure-oriented language which is also displayed as control blocks.

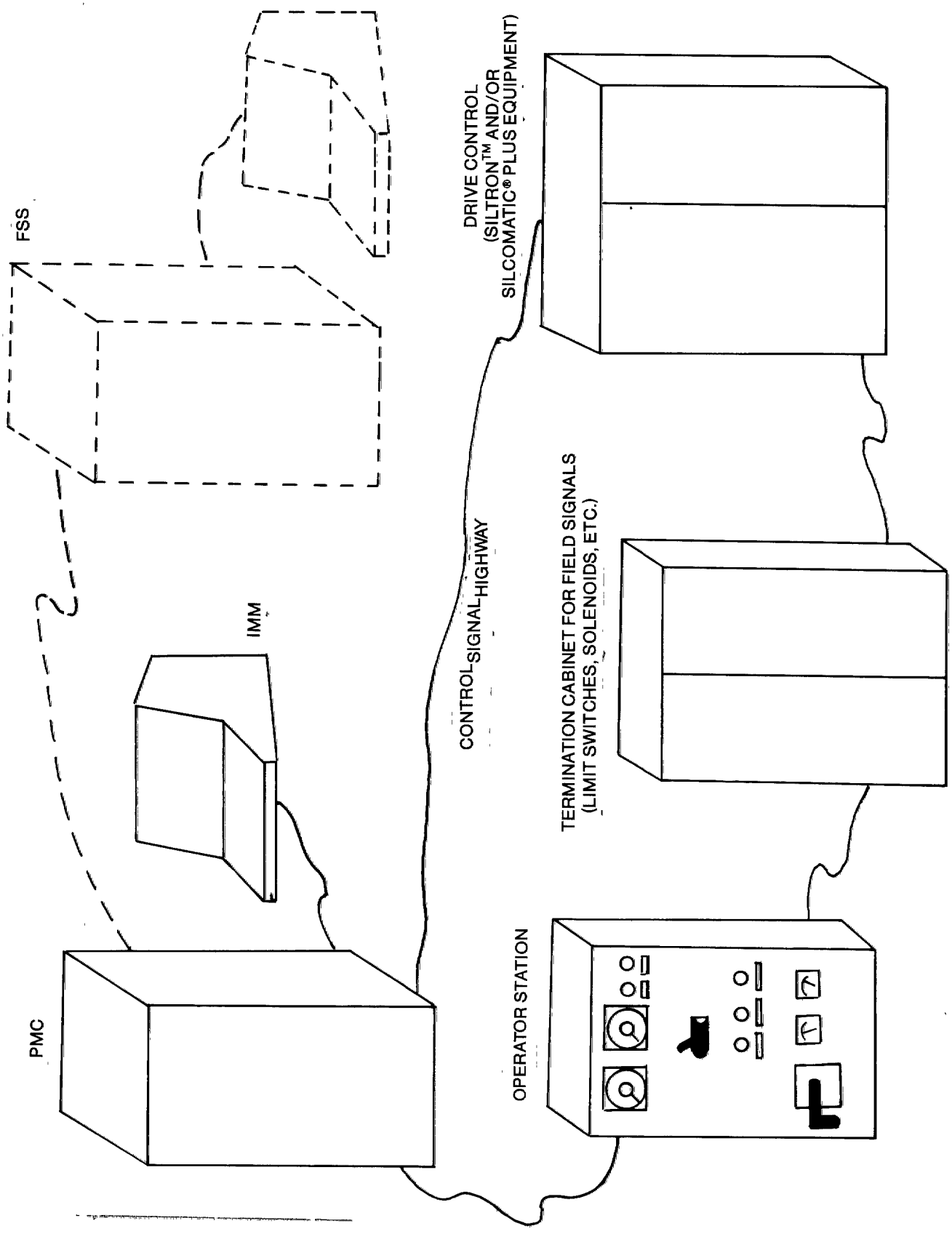


Figure 2-1.
Typical DIRECTO-MATIC® Plus System Configuration

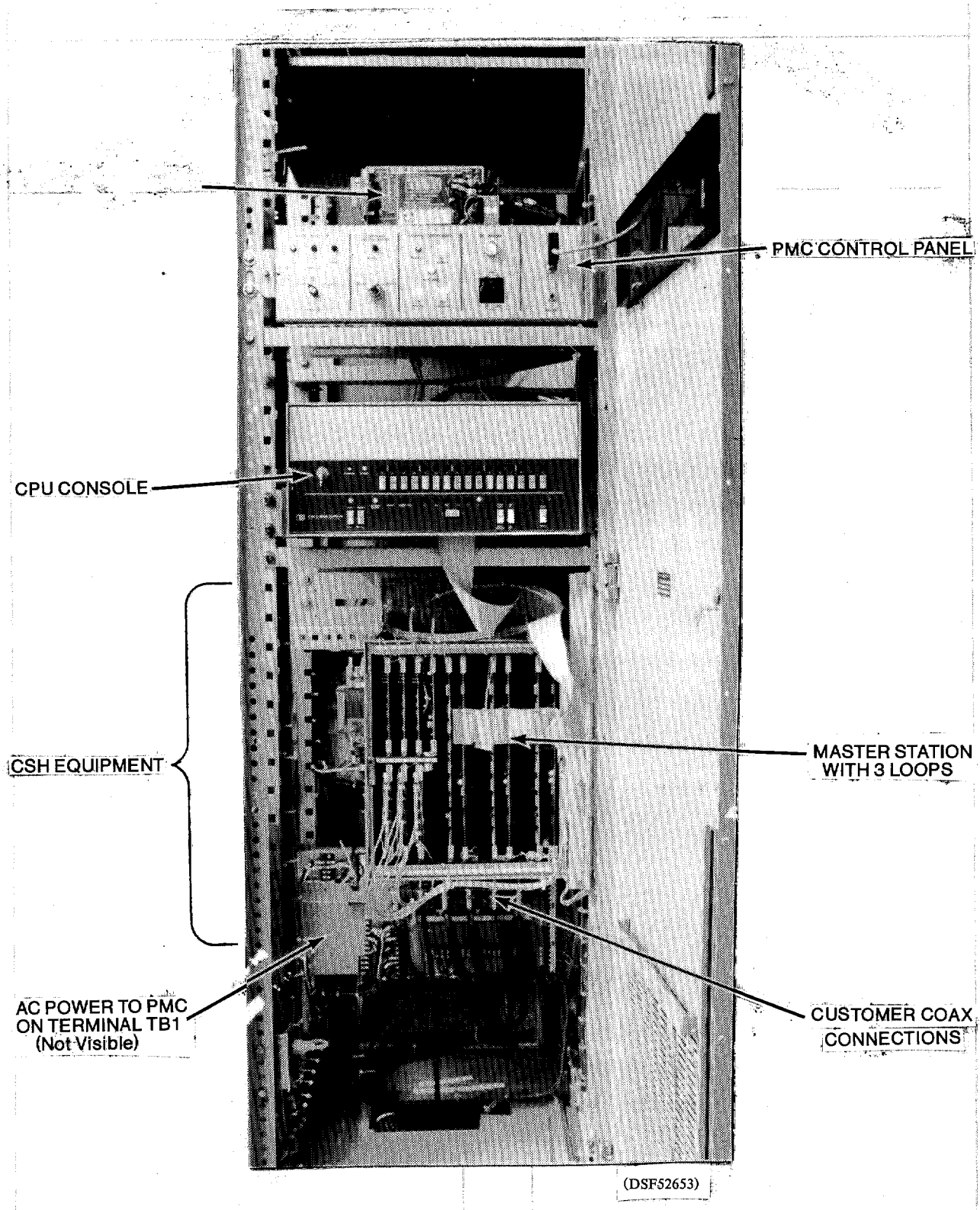


Figure 2-2.
PMC Equipment Cabinet

The particular functions performed by the PMC in a specific control system are indicated on the system elementary supplied with that system. This elementary diagram is in relay diagram form for sequencing functions or in control block form for regulating functions. The system elementary shows the interconnection of various control blocks using a signal number analogous to a wire number in the traditional hardware system. In some cases, a signal name also appears on the system elementary diagram.

The operation of the system can be changed by modifying the program within the PMC. The basic tool used to accomplish such changes is the Installation Maintenance Module (IMM). The IMM allows the display or modification of any control block within the system. The form of display used by the IMM is the relay diagram for relay functions and a "Control Block Image" form for the other type of control blocks. The block image form can be directly related to the control block of the system elementary by using the "Control Block Macro format" presented in Control Block Language Reference Manual GEK-65426.

An optional means of program modification is the Field Support System (FSS). The FSS consists of an HP-1000 computer with a removable disk containing control system information. The FSS is connected via a DS-1000 communications link to the PMC and allows the user to display and make modifications to any control block in the system using the pictorial format appearing on the system elementary. The FSS allows signal identification by use of the signal name appearing on the system elementary. Many additional functions are possible such as cross reference lists, trending printout, or interaction which enables the display of real time information from the PMC.

DESCRIPTION OF EQUIPMENT

The PMC equipment is housed in a metal cabinet with a hinged front door. Figure 2-2 shows the equipment complement of the cabinet. Two fans provide forced air ventilation through the bottom of the cabinet and up through the open grill at the top. Customer cables enter through the bottom of the cabinet and are connected to plugs below the electronics module.

PMC CONTROL PANEL

Figure 2-3 is an outline drawing of a PMC control panel. This panel is located near the top of the equipment cabinet and is used for monitoring the status of major system functions, selecting the operating mode and controlling PMC operation.

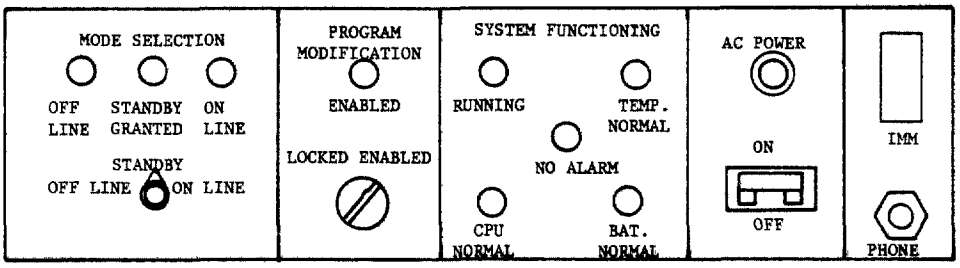


Figure 2-3.
PMC Control Panel

Mode Selection

There are three PMC modes of operation. The mode is operator selectable via the MODE SELECTION switch. The current operating mode is indicated by one of the three MODE SELECTION lights being lit. The specific functions of the operating modes for a particular system are shown on the system elementary. However, the general functions of the modes are as follows:

Mode	Associated Functions
OFF LINE	PMC operates internally but no control functions are executed.
STANDBY	PMC operates with only partial control functions. The control signal highway is active. Generally, all process inputs are being read by the PMC but outputs are sent only to display devices. Drives are not operated.
ON LINE	System completely operational. The drives are being given control signals over the control signal highway.

Program Modification

Modification of PMC memory (by IMM operator commands) may be permitted or inhibited by the PROGRAM MODIFICATION key switch which has the following positions:

Switch Position	Function
LOCKED	Inhibits any operator modification of PMC MEMORY.
ENABLED	Permits operator modification of PMC memory within constraints of IMM operator commands.

An ENABLED light above the keyswitch indicates the current status of the keyswitch by coming on when the keyswitch is in the ENABLED position.

System Functioning

Five lights are mounted on the SYSTEM FUNCTIONING panel of the PMC. These lights indicate the current operating status of the PMC as follows:

Indicator	Meaning When Lit
RUNNING	The PMC control executive program is running. If this light goes off, the system is stalled.
TEMP NORMAL	The temperature within the Central Processing Unit (CPU) cabinet is within the normal range.
CPU NORMAL	Voltages within the CPU are normal.
BAT NORMAL	Not Applicable.
NO ALARM	No alarm condition(s) currently exists. If the NO ALARM light goes off, an alarm is indicated on the PMC control panel. Depressing this lighted pushbutton will cause all existing alarm conditions to be cleared.

AC Power

The AC POWER switch, located on the PMC Control Panel, controls the AC power input to the PMC. A light above the switch, when lit, indicates the presence of AC input power. AC power input appears on TB1-5,6 whose location is shown in Figure 2-2.

Phone

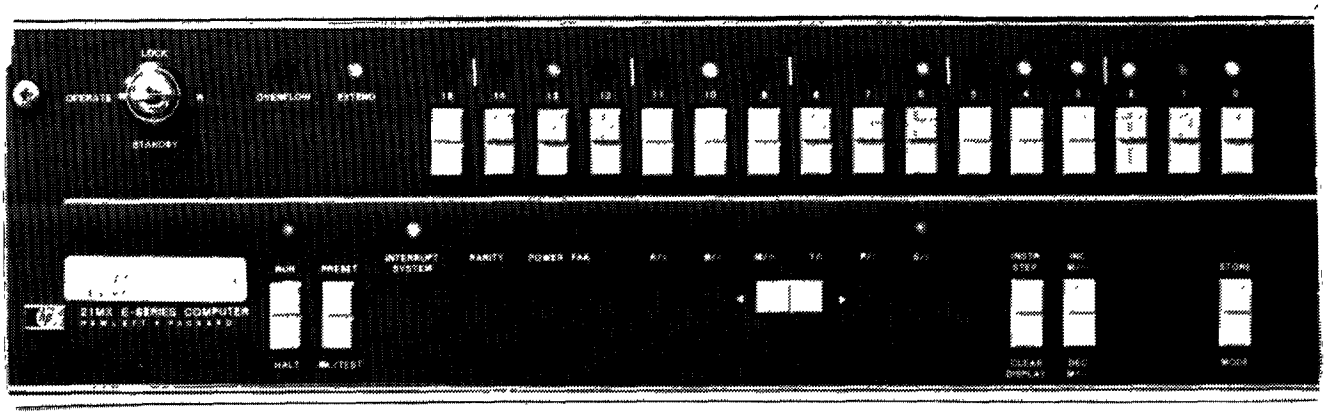
The PHONE jack is available for convenience when the PMC has been wired into a local intercom system.

IMM

The IMM connector accommodates the IMM CRT cable

CENTRAL PROCESSING UNIT (CPU) CONSOLE

The CPU console is shown in Figure 2-4. This console is located inside the PMC enclosure and displays system diagnostic information. The CPU console is also used for independent checkout and operation of the HP1000 processor (Refer to Hewlett-Packard 21MX E-Series Computer Installation and Service Manual).

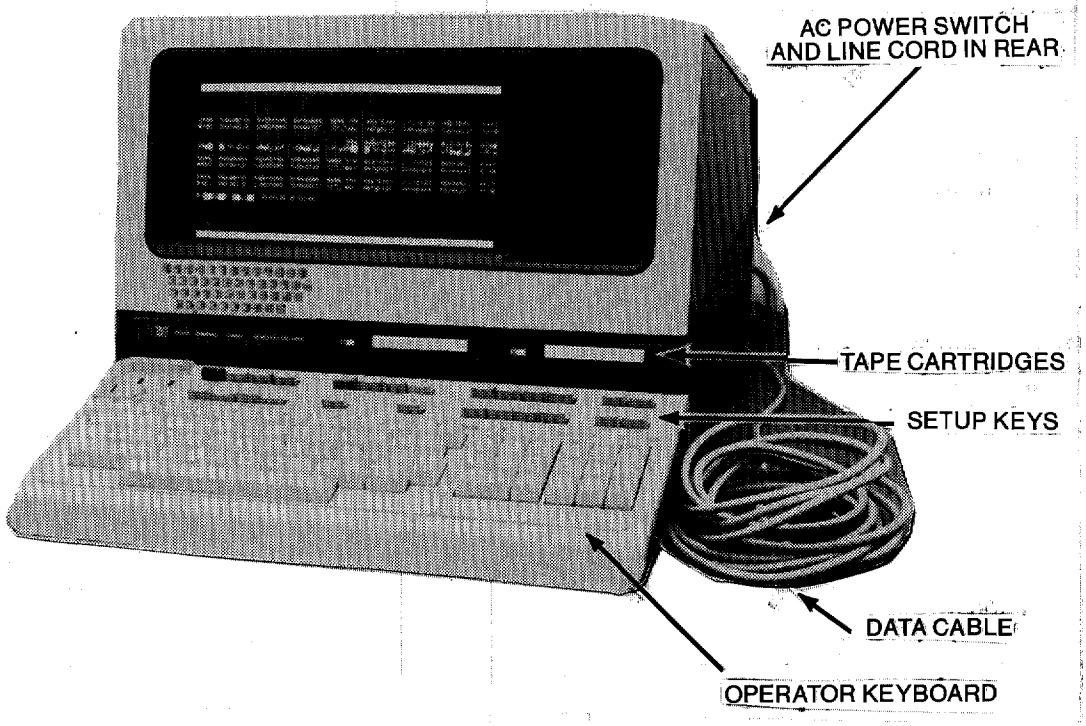


(DSF-52777)

Figure 2-4.
CPU Console

INSTALLATION MAINTENANCE MODULE (IMM)

The IMM is a special form of the HP2645 terminal containing General Electric software which accommodates control block language and ladder diagrams. The IMM is used with the PMC system and FSS (See Figure 2-5).



(DSF53593)

Figure 2-5.
IMM Front View

IMM FUNCTIONS

The IMM contains a terminal keyboard with CRT display and cartridge tape drive. The user enters commands via the IMM keyboard and views the results on the CRT screen. Information contained on an IMM cartridge tape can be loaded into the PMC, or information from the PMC can be copied onto the IMM cartridge tape. Both the load and copy functions are performed under the manual control of the IMM terminal keyboard.

The IMM allows the user to perform the following major functions:

1. Load, copy and compare load media. (Refer to **System Loading and Startup** and **Normal IMM Operating Procedure** sections for instructions).

2. Modify a control block system. In response to user commands, the IMM changes control blocks previously loaded into the PMC. (Refer to section describing **Normal IMM Operating Procedure** for details).

3. Monitor and/or alter the contents of the PMC. Under control of the IMM, PMC data can be monitored or changed. (Refer to **IMM User Interface Instructions** section for details).

4. Monitor the on-line status of control signals as they apply to the process controlled. (Refer to instructions describing the use of the DISPV and DSPLD commands in Section 3).

SECTION 3
OPERATING INSTRUCTIONS

The instructions in this section cover the following procedures:

- System Loading and Start up (Cold Start up)
- System Start up, Program Loaded (Warm Start up)
- System Program Modification
- System Diagnostic Display
- Normal Shutdown Procedure

WARNING

ENSURE THAT ALL DRIVES AND DEVICES ARE DISABLED FROM RESPONDING TO SYSTEM COMMANDS RESULTING FROM THE FOLLOWING PROCEDURES. OTHERWISE, PERSONNEL NEARBY THE PROCESS COULD BE INJURED AND EQUIPMENT COULD BE DAMAGED.

SYSTEM LOADING AND STARTUP (COLD SYSTEM)

The following start up procedure is used on a cold system, i.e., one which has been without power for over approximately 1.6 hours (assuming a good battery) or has been down for hardware maintenance. This procedure is also used when a new control program is to be loaded. It is assumed that all equipment is turned off. The procedures are given in the following order:

- PMC Power up
- IMM Power up
- PMC Loading - General
- IMM Loading via Cassette
- FSS Loading
- System Start up
- Set Time and Date

PMC Power up

1. Set CPU POWER switch (behind front panel) to OFF.
2. Set BATTERY PACK EXT-INT-OFF switch (on rear of CPU) to OFF.
3. Set AC POWER switch on PMC control panel to ON.
4. Set PS-5 power supply switch CB1 to ON.
5. Set CPU POWER switch to ON.
6. Wait 10 seconds for self-test to complete.
7. Set BATTERY PACK EXT-INT-OFF switch to INT.

IMM Power up

1. Check that the IMM date connector is plugged into the IMM outlet located on the front of the PMC control panel.
2. Check that ~ LINE switch (at rear of IMM CRT console) is set to ON position.
3. Set IMM console REMOTE and CAPS LOCK keys to DOWN position.
4. Set DUPLEX switch to FULL (down position).
5. Set PARITY Switch to NONE (down position).
6. Set BAUD RATE Selector switch to 9600 position.
7. Check that the POWER ON switch, located on the back of the IMM console, is set to ON position.

PMC Loading

The PMC object code generated by the FSS translator is available on up to three cassette tapes:

Tape 1 contains the first 32K of PMC memory. This includes the operating system and all PMC data tables. The data tables are in an area of memory called Subsystem Global Area (SSGA). Included in the SSGA are the following types of data:

1. System global constant values
2. System I/O variable values
3. Segment Status blocks for segments
4. Literal values for control blocks
5. Data formats
6. CSH I/O control tables

Tape 2 Contains part 1 of the remainder of PMC memory (over 32K). This tape consists only of the following partitioned programs (See Appendix G for abbreviations):

1. IMM (support program for operator interaction).
2. FSSCM
3. COMCM
4. OUTPG (text output)
5. MODE (system initialization)
6. CIAD
7. PRIV1 through $PRIV_n$

Tape 3 contains a continuation of tape 2 for the remainder of PMC memory (over 32K).

If a Field Support System (FSS) is connected to the PMC, the PMC object code will also be available in files on the FSS disk. The first 32K (Operating system and the SSGA which were on Tape 1) will be in a file named PONNNN, where NNNN is the PMC system ID number. The PMC partitioned programs (which were on tapes 2 and 3) will be in individual files named AONNN, A1NNN, A2NNN, etc. If an FSS computer is present, it will be connected to the PMC by a DS-1000 data link system, and the entire PMC software system may be "down-line" loaded from the FSS into the PMC memory.

NOTE

PMC loading must be accomplished by one of two methods: (1) via IMM cassette or (2) via FSS.

IMM Loading via Cassette

1. Load tape 1 by use of the PMC loader ROM. (See Appendix A for step-by-step instructions).
2. Initialize the PMC. (See Appendix B for step-by-step instructions).
3. Load tape 2 as follows:
    a. Insert cassette #2 into one of the two tape drives (left or right).
    b. Depress any key on the keyboard. A prompt character (*) will be displayed.
    c. Enter LO, N © (N = 4 if cassette is in left drive, N = 5 if cassette is in right drive; © = carriage return.

The first partitioned program on the tape will be loaded into PMC memory. After the first program has finished loading, the following message appears on the IMM CRT screen:

APLDR : DONE - IMM

Insert cassette No. 3 and repeat steps (b) and (c) for partitioned programs on tape 3.

NOTE

If the PMC was loaded via cassette tape, proceed to system start up procedure.

FSS Loading.

This procedure describes how to load the PMC from FSS disk files.

1. Load file PONNNN (where NNNN = system ID number) using the PMC down-Line loader ROM. (See Appendix C for step-by-step instructions).
2. Load the partitioned programs from the FSS terminal by using the FLOAD command in the FSS Operator Interaction program. (See FSS Operator Manual GEK-65435 for FLOAD command instructions).
3. An automatic START UP command will be performed. The following message will appear on the IMM CRT screen:

NNNN (Control System Title)

SYSTEM HAS BEEN BOOTED

Where NNNN = Control System ID

NOTE

The above procedure results in automatic system start up.

System Start up

NOTE

If cassette tapes are used, perform the following procedure for system start up.

NOTE

In the following narrative, command names keyed into the IMM are enclosed in a box.

1. Get the attention of the IMM Command Listener as described in IMM User Interface Instructions.
2. Perform system start up by entering the following command:

STRTUP ©

The system start up command will set the system booted flag if the privileged segments have been loaded. The privileged segments include the partitioned programs $PRIV_1$ thru $PRIV_N$ containing continuous initialize and interrupt segments. At this point, the following message appears on the IMM CRT screen:

NNNN SYSTEM HAS BEEN BOOTED
(NNNN = control system ID)

The "system booted" flag must be set in order to put the system in a **STANDBY** or **ON LINE** mode.

Set Time and Date

This command is used to enter system time, and optionally the date into the PMC. If a date is entered, only the first three characters of the month are used. (See **IMM User Interface Instructions** section for general IMM operating instructions)

Command Format:

STIME, hours, minutes, seconds © STIME, hours, minutes, seconds, day, month, year ©

Example:

STIME, 15, 52, 06, 3, DEC, 1979 ©

The PMC time is set to 15:52 and the date to 3 December 1979.

SYSTEM STARTUP, PROGRAM LOADED (WARM START UP)

In the event of a line power failure that lasts less than approximately 1.6 hours (assuming a good battery) the system will automatically return to operating status when line power is restored.

In the event that the system is manually turned off for less than approximately 1.6 hours, it may be returned to operation by setting the MODE SELECTION switch to the ON LINE position.

SYSTEM PROGRAM MODIFICATION

System programs can be modified by use of either the IMM or the FSS. In either case, the control blocks are altered to reflect the desired system configuration. The system elementary diagram of the system to be modified shows the entire configuration of control blocks and their interconnections. Control block functions are described in Control Block Language Reference Manual GEK-65426. If the FSS is used to modify the system, the modification is accomplished using control block diagrams as they appear in the reference manual. FSS operating instructions are given in the Field Support System Operator Manual GEK-65435.

When the IMM is used to modify a system, the program is manipulated by making changes to the control blocks similar to the modifications made with the FSS. IMM operations provide for the use of a relay diagram format for the modification of sequencing and interlocking control blocks. However, analog or count blocks are not presented in picture form on the IMM as they are on the FSS. Rather, they are presented in a "control block image" form using selectable hexadecimal or octal codes which relates directly to the CBL macro instruction format described in the CBL Reference Manual. The control block is identified by the block number, segment, and drive number as it appears on the FSS and system elementary. The control block parameters are indicated by their relative position as shown in the CBL macro instruction format.

It is necessary to correlate information from a system elementary with the related CBL instruction format which is displayed on an IMM CRT in order to modify a drive system program. The following is a simple example of a typical system modification using an IMM. (Refer to Figure 3-1).

1. The system elementary for segment 300A contains control block 420 (type ASUM) with a JOG signal input whose source (signal number) is 300A280. The JOG signal is presently the third input to ASUM block 420 (input B, referring to the control block photograph shown in Figure 3-1).

2. The user desires to replace the JOG input with an OFFSET input whose signal number is 200A370, i.e., the output of block 370. This replacement is accomplished by entering a DSPCB (display control block) command on the IMM. Either an octal or hex display may be displayed as shown in Figure 3-1. The resulting IMM readout for block 420 shows the address codes actually used in the control block and corresponds to the control block macro for an ASUM block as indicated in Figure 3-1.

3. Inputs 1, 2, and 3 of the system elementary control block correspond to positions 2, 3, and 4 of the ASUM control block macro. Therefore, position 4 of the ASUM block corresponds to the third input (JOG input) as it appears on the system elementary. The hex address code equivalent (IMM readout) is 3C2A. Thus, the signal name (JOG), signal number 280, and hex address code 3C2A are equivalent.

4. The DSPCB command is then used with control block 370. The output signal will be shown as the last address code (corresponding to output C, position 5 - refer to CBL Reference Manual). Assume this hex address code is 3F20 (establishing equivalency to signal name OFFSET, signal number 300A370).

5. Set the PMC control panel PROGRAM MODIFICATION key switch to the ENABLED position.

6. The segment to be modified must be disabled by entering the code "D" (See CMOD command in **Controller Miscellaneous Commands** section).

7. To replace the JOG signal with the OFFSET signal (control block 420 must be modified), the MDFCB (modify control block) command is entered into the IMM. The POS 4 hex address code 3C2A (signal number 280) is now replaced by hex address code 3F20 (signal number 370).

8. Re-enable the segment by entering code "E" (See CMOD command in **Controller Miscellaneous Commands** section).

9. The JOG input of the ASUM control block has now been replaced by the OFFSET input.

Further, assume the user desires to change the analog constant #22.1, first input of ASUM block 420, input data R (located at macro hex address code 40AD):

Enter CHGABS (change absolute) command into IMM as follows:

CHGABS, 40AD, A, #22.1 ©

IMM User Interface Instructions

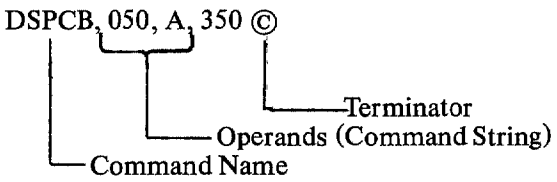
The IMM is used by first enabling the "IMM Command Listener". Then specific IMM commands are entered. This procedure and the various IMM commands are described below.

IMM Command Listener. The IMM command entry procedure is comprised of three steps:

1. Press any key on the console and an asterisk (*) prompt will appear.
2. Type in RU, IMM followed by a carriage return. The message ENTER COMMAND will appear on the CRT screen.
3. Enter desired command.

IMM Command Format. All the IMM functions are initiated by entering a command via the terminal keyboard. A command input is expected whenever the prompt ENTER COMMAND is displayed on the CRT. As a command is being keyed in, it will appear character-by-character on the next line of the CRT. When the command has been completely keyed in, a terminate action is required which signals the IMM to interpret the command and carry out the requested function.

All commands have the following general format:.



Command Name
- Alphabetic
- Identifies function

Command Operands
- Alpha numeric
- Separataed by comma(s),
- Variable number of operands
- Indicates a choice of parameters

Terminator
- Carriage return key,
- Designated by ©

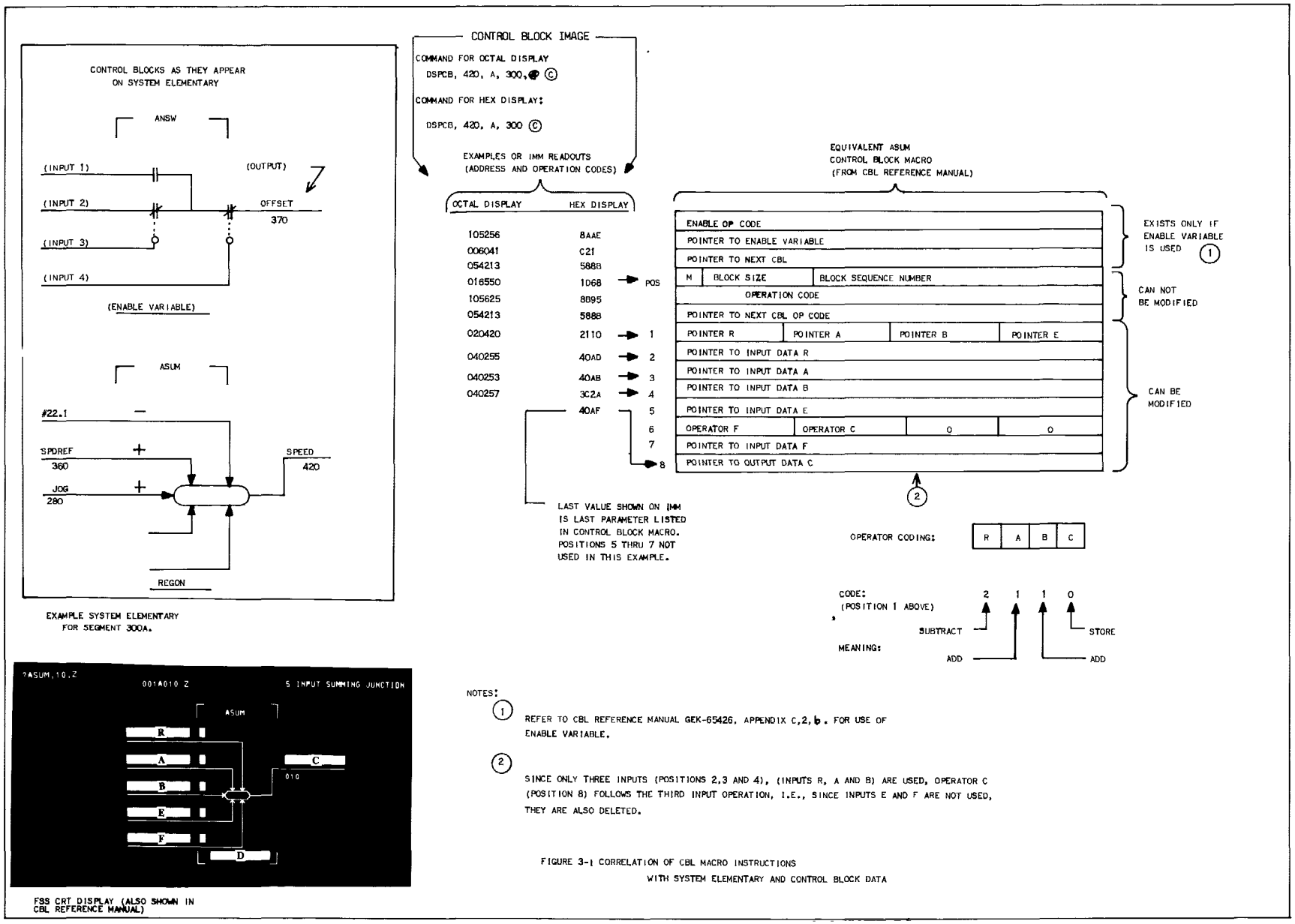


Figure 3-1.
Correlation of CBL Macro Instructions With System Elementary and Control Block Data

Commands shown in this manual follow certain conventions which are illustrated below:

DSPCB, blk, seg, drive ©

- The command name, shown in upper case letters, must be entered exactly as shown.
- The operands are variables shown in lower case letters. They are to be replaced by values as defined in the text.
- There is no comma between the last operand and the terminator.
- Operand leading and trailing blanks are ignored.
- Optional operand parameters are shown in brackets, e.g.



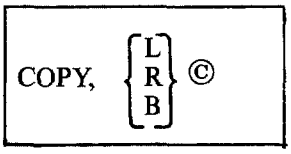
Controller Copy/Compare Commands

Copy Controller. This command copies the SSGA and all partitioned programs from the PMC memory onto casette tape. The SSGA is copied onto one cassette and the partitioned programs onto another one or two others. (A second tape is needed if the first is filled up. The number of programs written on a tape may not be the same as a WTAPE command of the FSS).

The SSGA is copied in a format acceptable to the cassette loader ROM. When reloading a PMC system, the original FSS translator output (See description of "Tape 1" in PMC Loading section) must be loaded first followed by the SSGA copy tape.

The partitioned programs are copied in a format acceptable to the PMC "LO" command. (See description of "tape 2" in PMC Loading section).

Command Format:



L = Left cassette drive
R = Right cassette drive
B = Both cassette drives

The SSGA tape is always copied first. If an "L" or "R" were entered, at the completion of the SSGA copy, a message is output to the CRT. This message directs the operator to install a new tape in the specified drive and to type in "GO © " to start the copy of tape 2.

If "B" were entered, the SSGA would be copied on the left drive cassette and partitioned programs on the right drive cassette. If still another tape is needed, the operator will be informed to insert another tape and to tell the system which tape unit is to be used (L or R).

Compare Controller. This command is used to compare the tapes (SSGA tape and partitioned program tapes) produced by a previous COPY command from the corresponding areas in PMC memory. Non-compares are displayed on the CRT in hexadecimal format, as shown below. Up to 80 non-compares will be displayed plus a count of total non-compares.

ADDR	MEM	TAPE
AAAA	MMMM	TTTT
"	"	"
"	"	"
"	"	"

——— up to 20 lines ———

Where: AAAA = hex address
MMMM = memory contents
TTTT = tape contents

Command Format:

COMPAR, {L / B / R} ©

COMPAR, {L / B / R} Beg. address End. address ©

COMPAR, {L / B / R} Partitioned program name ©

L = Left cassette drive
R = Right cassette drive
B = Both cassette drives

If it is desired to compare only a particular area within the SSGA, enter the beginning and ending addresses of the area.

If it is desired to compare only one particular partitioned program on tape, enter the program or segment name.

CONTROL BLOCK EDIT COMMANDS

Parameters for Edit Commands. The following command parameters are applicable to the Control Block Edit Commands discussed in this section:

blk	Control block sequence number (always a decimal integer) 001 ≤ NNN ≤ 999
seg	Segment identifier (alpha) A to Z
dr	Drive identifier (decimal integer) 001 ≤ NNN ≤ 999
n	Number of words of data (decimal integer) 03 ≤ NN ≤ 31. For LGIC, LRLY, LGTT, INZS, INZO, 03 ≤ NN ≤ 62.
data	A word entry in a control block. 0000 ≤ HHHH ≤ FFFF, (hexadecimal) @000000 ≤ NNNNNN ≤ @177777, (octal)
pos	Position in a control block template relative to the "NEXT CONTROL BLOCK POINTER" taken as 0 (decimal integer).
type	H hexadecimal format @ octal format

Display Control Block.

The following command is used to display the code for a specified control block. The display code corresponds to the CBL macro formats catalogued in the CBL Reference Manual.

Command Format:

```
DSPCB, blk, seg, dr ©
DSPCB, blk, seg, dr, type ©
```

Example:

```
DSPCB, 040, B, 350 ©   Command

HHHH
HHHH
HHHH
HHHH
HHHH
```

Resulting CRT picture

Default for "type" is hexadecimal.

Insert Control Block.

Used to insert a control block in place of an existing control block, or between existing control blocks.

The control block sequence number, segment and drive must be specified. The number of words of data and the hexadecimal or octal data for each must be provided.

The first word and "next control block" pointer in the new block may be entered as "0" (or any other value). It will be properly filled in internally by the command.

The code for the inserted block is physically located in the segment spare area.

Command Format:

INSCB, blk, seg, dr, n, $data_1$, . . .

Example:

INSCB, 210, B, 250, 5, data1, data2, data3, data4, data5 ©

Control block 210 is inserted into segment 250B. "Next control block" pointers are automatically adjusted.

Modify Control Block.

Used to modify an existing control block. Block sequence number, segment and drive must be specified. A single word in the block can be changed with each command entry. An integer word position indicator, relative to zero for the "Pointer to Next Control Block", is required. Data format is hexadecimal or octal.

Block sequence number, op code, and "Pointer to Next Control Block" can not be changed.

Command Format:

MDFCB, blk, seg, dr, pos, data ©

Example:

MDFCB, 210, A 250, 3, 1430 © Command

The third word relative to "pointer to next CB" of block 210, segment 250A, is changed to 1430 hexadecimal.

Delete Control Block.

Used to delete a control block from the system.

Control block sequence number, segment and drive must be identified.

"Next Control Block Pointer" of the preceding block is automatically adjusted.

Command Format:

DLTCB, blk, seg, dr ©

Example:

DLTCB, 040, B, 350, © Command

Control block 040 of segment 350B is bypassed by changing the next block pointer in the preceding block.

Display Relay Ladder Control Block.

Used to display a relay ladder control block with current status of all signals in the control block.

The control block sequence number, segment and drive must be specified.

The current status may be optionally trended by specifying a number of samples and sample rate. A positive repetition rate is interpreted as samples per second; a negative value as seconds between samples.

A contact or coil which is active (i.e., current is flowing through it) is displayed on the scren in inverse video.

Command Format:

```
DSPLD, blk, seg, dr ©
DSPLD, blk, seg, dr, # samples ©
DSPLD, blk, seg, dr, # samples, rep rate ©
```

Example:

DSPLD, 140, F, 500, 10, -2 ©

The appropriate relay ladder will be displayed on the CRT with signal status being updated 10 times at 2-second intervals.

Default values for samples and repetition rate are one sample and once per second, respectively.

Insert Relay Ladder Control Block.

Used to insert a new relay ladder control block between existing control blocks in a segment.

The new control block sequence number, segment and drive must be specified. The signal identification to be entered during "label" editing is the memory address of the signal value.

The macro for the inserted block will be physically located in the segment's "CBL spare" which was declared at the time the segment ID was specified. If a control block requires "internal" variables, i.e., addresses within the macro which cannot be specified via the relay ladder diagram, they must be entered with the MDFCB command.

Command Format:

```
INSLD, CB name, blk, seg, dr  E
```

Example:

INSLD, KRLY, 210, B, 300 E Command

The relay ladder template will be displayed for editing.

Modify Relay Ladder Control Block.

Used to modify an existing relay ladder control block.

The control block sequence number, segment and drive must be specified. If signal identification is changed, it may only be specified by memory addresses of the signal values.

If the size of the control block macro generated from the modified relay ladder is less than or equal to the size of the original control block, the original block will be overlayed with the modified block. If the size of the control block macro generated from the modified relay ladder is greater than the size of the original control block, the modified block will be put in the segment's "CBL spare" which was declared at the time the segment ID was specified.

Command Format:

MDFLD, blk, seg, dr E

Example:

MDFLD, 210, A, 250 E

The relay ladder control block will be displayed for editing.

Delete Relay Ladder Control Block.

Used to delete a relay ladder control block from the system.

Control block sequence number, segment and drive must be specified.

The "Next control block pointer" of the preceding block is adjusted to bypass execution of this block.

Command Format:

DLTLD, blk, seg, dr E

Example:

DLTLD, 040, B, 350 E Command

Control block 040 of segment 350B is bypassed by changing the "next block pointer" in the preceding block.

CONTROLLER MONITOR/CHANGE COMMANDS

Parameters for Monitor/Change Commands.

The following parameters apply to the Controller Monitor/Change Commands following:

blk	Control block sequence number (dec. integer) 001 ≤ NNN ≤ 999
seg	Segment identifier (alpha) A to Z
dr	Drive identifier (dec. integer) 001 ≤ NNN ≤ 999
addr	Memory address (hexadecimal or octal) 000 ≤ NNNN ≤ 7FFF (hexadecimal) @000000 ≤ NNNNNN ≤ @077777 (octal)

value	Contents of a signal or memory word (specified by TYPE parameters).
type	Format of a signal or memory word (alpha) A - analog (1×10^{-38} to 1×10^{38}) C - count (-32768 to 32767) L - logic (= T or F) T - text H - hexadecimal @ - octal
pos	Position in a control block code template relative to 0 for the "Next Control Block Pointer" (decimal integer) $1 \leqslant NN \leqslant 59$
# lines	Number of display lines on the CRT (decimal integer)
rep rate	Repetition rate (decimal integer) +NN - samples per second -NN - seconds between samples
use	Signal type or use (alpha) I - input O - output
device	Output device identifier (dec. integer) 1 - CRT 6 - line printer Default device is the CRT.
page	Extended memory page number (hexadecimal or octal) $00 \leqslant NN \leqslant 3F$, hexadecimal $@000 \leqslant @NNN \leqslant @177$, octal
index	Offset into an extended memory page (hexadecimal or octal) $000 \leqslant NNN \leqslant 3FF$, hexadecimal $@0000 \leqslant NNNN \leqslant @1777$, octal
# words	Number of consecutive memory words (hexadecimal or octal)
percent	+ or - decimal percentage of desired increment (decimal integer)
channel	Trend record channel number (dec. integer) $1 \leqslant N \leqslant 6$
slope	Ranging gain for trending (analog value) $0.001 \leqslant NNNNN \leqslant 99999$
offset	Ranging offset for trending (analog value) $-99999 \leqslant NNNNN \leqslant 99999$

Display Values.

Used to display memory contents from the PMC, on the IMM console CRT. Up to six values can be displayed at one time. For each value, a memory address and signal type must be entered via a single command entry.

The number of lines and repetition rate must be specified. A positive repetition rate is interpreted as samples per second. A negative value indicates seconds between samples.

Format:

DISPV , # lines, rep rate, addr 1, type 1, addr 2, type 2,...addr n, type n ©

Example:

DISPV, 4, -10, 13A8, A, 1E77, C © Command

13A8	1E77	
380.7	56	CRT Display
382.1	56	
386.2	56	
390.5	56	

Four lines are displayed at 10 second intervals.

Dump Memory

Used to dump a block of SSGA (Subsystem Global Area) in PMC memory to a printing device. Memory beginning and ending addresses must be specified.

Data type may be specified for output formatting. Default is hexadecimal. (Data type "L" will be treated same as "C").

An output device may be specified. If not, output is to the CRT.

Format:

DUMP, beg addr, end addr ©
DUMP, beg addr, end addr, type ©
DUMP, beg addr, end addr, type, device ©

Example:

DUMP, 3A00, 3A20, H, 1 © Command

address	data (1)...data (8)	
"	" "	Resulting dump on CRT, Values in counts.
"	" "	
"	" "	
"	" "	

Dump Extended Memory

Used to dump information from PMC memory above the lower 32K words to a printing device.

Data is addressed by specifying hex or octal memory "page" and an "index" within the page. Page, index, or words preceded by @ indicates an octal designation.

Data type may be specified for output formatting. Default is hexadecimal. (Data type "L" will be treated the same as "C").

An output device may be specified. If not, output is to the IMM console CRT.

Format:

DMPEM, page, index, # words ©
DMPEM, page, index, # words, type ©
DMPEM, page, index, # words, type, device ©

Example:

DMPEM, 1, @100, @23, C, 1 © Command

PAGE = @00001 OFFSET = @000100

000100	data (1) ... data (8)	Resulting dump on CRT, values in counts
000110	" "	
000120	" "	
000130	" "	
000140	" "	

Dump Segment.

Used to dump a given portion of a segment to a printing device. Beginning and ending addresses must be specified.

Data type may be specified for output formatting. Default is hexadecimal. (Data type "L" will be treated as "C").

An output device may be specified. If not, output is to the CRT.

Format:

DMPSEG, dr, seg, beg addr, end addr ©
DMPSEG, DR, SEG, BEG ADDR, END ADDR, TYPE ©
DMPSEG, dr, seg, beg addr, end addr, type, device ©

Example:

DMPSEG, 300, A, 5400, 5410 © Command

5400	data(1).....data(8)	Resulting dump on CRT
5408	" "	
"	" "	
"	" "	

Change Memory Absolute.

Used to change the value of a memory location in the PMC I/O tables (SSGA).

Memory address and signal type must be given.

The old and new values will be printed out.

Format:

CHGABS, addr, type, value ©

Example:

CHGABS, 3A2B, C, 515 © Command

If the given address, 3A2B, is within limits, its contents will be changed to 515 counts.

Change Memory Incremental.

Used to change the value of an analog or count value in PMC I/O tables by a specified percentage.

Memory address, signal type and a percentage value must be input.

The signal value may be increased or decreased by using a positive or negative percentage change.

The old and new values will be printed out.

Format:

CHGINC, addr, type, percent ©

Example:

CHGINC, 21FB, A, 50 © Command

If the given address, 21FB, is within limits, its contents will be increased by 50%.

Change Local Common Absolute.

Used to change the value of a memory location in the local common area of a segment (local common is the name applied to the block of data created with a PDATV control block).

Logical address within the segment and signal type must be given.

The old and new values will be printed out.

Format:

CHGLAB, dr, seg, addr, type, percent ©

Example:

CHGLAB, 300, A, 5410, C, 120 © Command

If the given address, 5410, is within limits, its contents will be changed to 120.

Change Local Common Incremental.

Used to change the value of a memory location in the local common area of a segment by a specified percentage (local common is the name applied to the block of data created with a PDATV control block).

The logical address within the segment, signal type and a percentage value must be given.

The signal value may be increased or decreased by using a positive or negative percentage change.

The old and new values will be printed out.

Format:

CHGLIN, dr, seg, addr, type, percent ©

Example:

CHGLIN, 300, A, 5410,C, 10 ©Command

If the given address, 5410, is within limits, its contents will be increased by 10%.

Trend Signals.

Used to trend signals from the PMC onto a strip chart recorder. The output to the recorder is accomplished via a "trend segment" which must be entered with the system at translation time. The control block diagram for channel 1 is shown in Appendix D. The set of control blocks shown would have to be duplicated for each channel, up to 12, on the recorder.

Recorder channel, signal address and signal type must be entered. Analog, Count and Logic are the allowed types.

Ranging of the signal can be accomplished by specifying a Slope and Offset.

Trend value = (SLOPE X SIGNAL) + OFFSET

Repetition rate, Slope and Offset are optional parameters. Default values are one sample per second, 1.0 and 0.0, respectively. Repetition rate applies to all active channels (i.e., the most recent repetition rate entered will be rate at which output is made to the recorder).

Trending on any channel is terminated by entering a command with a zero address.

Format:

```
TREND, channel, ©
TREND, channel, addr, type ©
TREND, channel, addr, type, rep rate, slope offset ©
```

Example 1:

TREND, 4, 2EA8, A, -5, 3.5, 0.0 © Command

The analog signal, from memory address 2EA8, will be trended on channel 4. The sample rate is changed to once every 5 seconds. A gain of 3.5 and an offset of 0.0 is applied to the signal before output.

Example 2:

TREND, 1, 0 © Command

Trending on channel 1 is terminated.

Force Signal.

Used to override or force signals and control block parameters in the PMC.

A signal or parameter to be forced is located by control block name, i.e., drive, segment and block number.

The position parameter identifies which control block signal or parameter is referenced. The Position parameter gives the position in the control block relative to 0 for the "Next Control Block" pointer.

Forcing can apply to a block input or output as indicated by the "Use" parameter.

Up to sixteen signals may be forced at one time. The FORCE command must be entered once for each signal.

Analog, Count and Logic signal types can be forced.

Any forced value can be changed by re-entering the FORCE command with a different new Value parameter.

Format:

FORCE, blk, seg, dr, pos, use, type, value ©

Example 1:

FORCE, 090, A, 350, 4, 0, A, 300.0 © Command

The output of block 350A090 is forced to a value of 300.0

Example 2:

FORCE, 200, A, 350, 2, I, L, T © Command

A logic input of block 350A200 is forced to the True state.

Cancel Force.

Used to remove forcing from a control block signal or parameter.

The control block and the signal position within the block template must be specified.

The original signal connection is restored.

Format:

CANCEL, blk, seg, dr, pos ©

Example:

CANCEL, 200, A, 350, 1 © Command

The block 350A200 input, reference by the parameter following the "Next Control Block Pointer", is changed back to its original connection.

Force/Cancel Relay Ladder Signals.

Used to temporarily force (override) and cancel force (remove override) on current status or values of signals in a relay ladder control block.

The control block sequence number, segment and drive must be specified. The specified relay ladder will be displayed with any currently forced signals indicated by half-bright inverse video.

Up to sixteen signals may be forced at the same time. Inputs are forced only for the specified block. Outputs are forced globally, i.e., a forced coil affects all blocks which reference it.

A logic signal is forced by positioning the cursor to the desired contact and typing "F". Typing "T" will cause the contact to toggle between open and closed. Inverse video indicates a closed state. A forced contact is indicated by half-bright inverse video of the signal name field. Typing "C" removes the force.

A count signal is forced by positioning the cursor to the desired value and typing "F", followed by the force value. A forced count signal is indicated by a half-bright count value. Typing "C" removes the force.

Format:

FRCLD, blk, seg, dr ©

Example:

FRCLD, 50, A, 100 © Command

The relay ladder control block will be displayed with any currently forced signals indicated by half-bright inverse video. Force or cancel-force as desired. Terminate the forcing/cancelling by pressing the "ENTER" pushbutton. The force/cancel-force requests will then be put into effect.

List Force Table.

This command will display all entries currently in the force table. The force table is used by both the FORCE and FRCLD commands.

Format:

```
LISTFR ©
```

Example:

LISTFR © Command

Resulting display on CRT:

SEG	BLK#	POS	USE	VALUE
100A	200	2	1	.12345E12
200B	10	4	0	32767
400V	110	9	0	T

Specify RCALL Control Block.

Used to specify the RCALL control block from which the base address is to be taken when signal addresses used in subsequent commands are for signals found within a Reusable Sequence. If CALLER is used with no parameters, the currently specified RCALL is cleared.

If bit 15 is set in a signal address, it indicates that this address has "extended address modification (EAM)." If signal addresses are obtained from a listing of the FSS Symbol File and the signal is from a Reusable Sequence segment, they may have EAM flags set in bits 12 through 15. These EAM flags indicate that the value given in bits 0 through 11 is an offset either into a data table (direct basing) or into a RCALL control block (indirect basing). Each user of the Reusable Sequence (RCALL CB's) has its own unique signals, internal and external, which are used by the Reusable Sequence. Further explanation of RCALL control blocks and Reusable Sequences may be found in the CBL Reference Manual GEK-65426.

Format:

```
CALLER, blk, seg, dr ©
CALLER ©
```

Example:

CALLER 50, A, 100 © Command

The RCALL control block, 100A050, will be used to obtain the base for any signal addresses which are based. This base remains in effect until a new RCALL is specified by a CALLER command or IMM is excited.

CONTROLLER MISCELLANEOUS COMMANDS

Parameters for Miscellaneous Commands

The following parameters apply to the Miscellaneous Commands following:

hour	Hour of the day (00 ≦ NN ≦ 23)
minute	Minute of the day (00 ≦ NN ≦ 59)
second	Second of the day (00 ≦ NN ≦ 59)
day	Day of month (00 ≦ NN ≦ 31)
month	First 3 characters of current month (AAA)
year	Current year (NNN)
drive	Drive number (000 ≦ NNN ≦ 999)
seg	Segment name (A to Z)
mode	Enable/Disable segment mode (E or D)

Display Mode

This command displays PMC system mode and, optionally, segment control mode.

The IMM must be connected to a PMC while using this function.

There are three PMC modes: OFF-LINE, STAND-BY, and ON-LINE.

There are two segment modes: DISABLED (D) and ENABLED (E).

Command options are provided for different mode displays:

- System mode plus modes of all segments in the system.
- System mode plus modes of all segments of a single drive.
- System mode plus the mode of a single segment.

Format:

```
DMOD ©
DMOD, drive ©
DMOD, drive, seg ©
```

Example 1:

DMOD, 350, C ©	Command
SYSTEM ON -LINE 350C-E	Resulting display

In the above example, segment C of drive 350 is enabled.

Example 2:

DMOD, 210 © Command

SYSTEM ON-LINE

210A-E 210B-E 210C-E. . . 210J-D
210L-D 21-M-D } Resulting display

Example 3:

DMOD Command

SYSTEM ON-LINE

100A-E 100 B-E 100C-E. . .100C-E
100M-E 100N-D

200A-E 200B-E 200C-D. . . 200J-D
200P-D. 200X-E

} Resulting display

Change Mode

This command is used to change the mode of a specified segment.

The segment can be Enabled or Disabled by entering the code "E" or "D", respectively.

Format:

CMOD, drive, seg, mode ©

Example:

CMOD, 350, D, E © Command

Segment 190D is enabled.

RELAY LADDER CONTROL BLOCK EDITING

Relay ladder control block editing is used whenever an "insert relay ladder" (INSLD) or "modify ladder" (MDFLD) command is requested for any of the following control blocks:

KRLY - Relay ladder array
KDLY - Time delay relay
KCTR - Counter
KHLD - Hold (early exit)
KLCH - Latch relay
KPLS - Pulse relay

Detailed functional descriptions of the above control blocks appear in Control Block Reference Manual GEK-65426.

This section contains instructions applicable to the following functions:

- General editing rules
- KRLY editing
- KCTR editing
- KDLY editing
- KLCH/KPLS/KHLD editing
- CRT self test
- Editing error messages.

General Editing Rules

Edit Modes. There are two edit modes. In either mode the cursor keys (arrows on IMM keyboard) are used to move the cursor to the circuit element or label to be edited. The field to be edited is indicated on the IMM CRT by inverse video. The two edit modes are:

1. EDIT CIRCUIT - add or delete circuit elements, e.g., contacts and connectors, or change the sense of contacts, normally open/normally closed.

2. EDIT LABELS - allows modification of the signal identification associated with a relay circuit element. The signal identification appears in an inverse video field. The form of the identification depends on whether the FSS or IMM is being used.

If the IMM is used, the field can contain one of two types of identification. If an INSLD or MDFLD command is executed, the signal is identified by the hex memory address code. However, if a DSPLD command is exercised and a CSH input is identified, it is expressed as a CSH address. If an internal signal is identified, it is expressed as a hex memory address code.

If the FSS is used, a signal name or signal number appears in the inverse video field.

The completion of editing is indicated by pressing the "ENTER" key. If the circuit and labels are all valid, the control block will be added to the system; otherwise an error message will be displayed. To assist the operator during editing, a menu of the editing keys is displayed on the bottom of the screen.

Edit Function Keys.

KEY	FUNCTION	DESCRIPTION
f1	ABORT	Abort editing session, no change in control system.
f2	LIST FUNCTIONS	Displays a menu of all function keys f1 - f8
f3	CHECK	Checks displayed KRLY block for circuit errors. If valid, the ladder is "normalized". If not valid, an error message is displayed.
f4	EDIT MODE	Switches the edit mode between "edit circuit" and "edit label".
f5	SELF TEST	Performs a self test on CRT RAM and ROM memory. Appropriate message is displayed.
f6	Unused	
f7	CLEAR ERROR	Clears displayed error message
f8	Unused	

Edit Cursor Keys. The following IMM terminal keys move the edit cursor from its present position to a new field. If the user attempts to move the cursor to a position beyond a boundary of the control block, a "wraparound" to the opposite boundary will occur.

KEY	DESCRIPTION
	Advances the edit cursor "up" one field
	Advances the edit cursor "down" one field
	Advances the edit cursor to the "right" one field
	Advances the edit cursor to the "left" one field
	Homes the edit cursor to the upper left hand corner of the relay ladder control block.
RETURN	Moves the edit cursor to the left-most field on current line

Edit Symbol Keys. The following keys alter the contents of the label field or circuit field pointed to by the edit cursor.

- Edit Label Mode

KEY	DESCRIPTION
A-Z,0-9	Depressing any of these keys will cause the corresponding character to be entered into the inverse video field. The cursor location shows the operator where the characters will be positioned within the field.
	Move the edit cursor back one position. Any displayed characters which are "backed over" will not be affected.
	Move the edit cursor to the right one character position. Any character "spaced over" will be blanked out.

- Edit Circuit Mode

KEY	*DESCRIPTION*
1	Place a normally open contact in the relay ladder.
2	Place a normally closed contact in the relay ladder
3	Place a horizontal short into the relay ladder.
4	Place a vertical short into the relay ladder.

Edit Delete Keys.

KEY	DESCRIPTION
DELETE CHAR.	Delete the characters in the label field or the relay element pointed to by the edit cursor, depending on mode. Advance the edit cursor right one position.
DELETE LINE	Delete both labels and relay elements on the relay ladder row pointed to by the edit cursor. Move the edit cursor to the first column in the deleted row.
CLEAR DISPLAY	Delete both labels and relay elements from the edit cursor position to the end of the relay ladder (lower right hand corner). The edit cursor position will not be changed.

Editing Complete Key.

KEY	DESCRIPTION
ENTER	The operator indicates editing is completed by pressing the "ENTER" key. If no errors are encountered, the edited control block is incorporated into the control system. If an error is encountered, a message is displayed. If the error condition is recoverable, the control block is automatically redisplayed and the edit mode entered.

KRLY Editing

Circuit Elements. The maximum relay ladder array possible is 5 rows by 7 columns for a total of 35 contact inputs with one coil output.

For editing purposes, the circuit is broken down into circuit elements consisting of:

- vertical connectors
- horizontal connectors
- normally open contacts
- normally closed contacts
- coil

Signal IDs. Each contact and the coil has a signal ID or "label" associated with it.

Normalization. When a "CHECK or "ENTER" is requested, the relay ladder is automatically "normalized", i.e., all circuit elements are repositioned toward the top and left to the maximum extent possible without changing the character of the circuit.

Circuit Wraparound. Within the 5X7 relay ladder diagram, it is possible to create relay strings with greater than 7 contacts in series. A maximum of 19 series contacts are possible. To extend a relay ladder row beyond 7 series contacts, the operator extends a relay row by connecting its end to the start of another row which is used as a shorting or "wraparound" row.

A maximum of two wraparounds are possible which yield 19 series contacts. However, in this configuration, no "OR-ing" of contacts is possible. The operator should be aware that the shorting row or wraparound row cannot be intersected.

A maximum of 5 OR'd relay ladder rungs is permitted. Note, however, that it is physically possible to create relay ladders with greater than five OR'd rungs within the 5X7 format. If the limit is exceeded, an error message will be displayed when normalization is attempted subsequent to pressing the "ENTER" or "CHECK" keys.

KCTR Editing

Circuit Elements. The Counter block has four contact inputs which may be "circuit edited" as follows:

CONTACT	KEY
Normally open	1
Normally closed	2
Horizontal short	3
Open connection	"DELETE CHAR"

Signal IDs. The signal IDs of the four contact inputs, coil output and three count signals may be "label edited".

KDLY Editing

Circuit Elements. Only the time delay value may be edited.

Signal IDs. The signal ID of the contact input and coil output may be "label edited".

KLCH/KPLS/KHLD Editing

Circuit Elements. No editing allowed on circuit elements.

Signal IDs. The signal IDs of the contact inputs and coil output may be "label edited".

CRT SELF-TEST

The CRT self test is activated by pressing function key "f5". This routine performs a RAM and PROM check. It outputs to the CRT the following information:

Program Revision Level XX

PROM NUMBER	PROM REV.	PASSED TEST?
1	X	Yes
2	X	Yes
.	.	.
.	.	.
N	X	Yes

RAM OK

The PROM number is a number assigned to each prom relating it to its card position. Prom REV is a number denoting the revision level of the PROM.

Passed test is answered YES or NO if the PROM agrees with its check sum.

The program revision level is the sum of the PROM revision level and tells the operator at a glance if all PROMS are the current revision level. To display the list of PROM revision levels on the IMM CRT, depress the RESET TERMINAL key twice in rapid succession to clear the terminal, then depress the following keys: ESC & C 70041 @ E The last entry is the RAM test. It will be either RAM OK" or "RAM Failed".

SYSTEM DIAGNOSTIC DISPLAYS

The diagnosis of many system faults is aided by use of the indicators on both the PMC control panel and the CPU console.

A PMC fault condition is indicated when the PMC control panel NO ALARM light is off. When this occurs, the 16 LEDs on the CPU console stop flickering and display a pattern whose configuration identifies some type of existing fault. The type of fault and appropriate operator actions are given in this section.

The CPU alarm indicators may be cleared by pressing the NO ALARM pushbutton on the PMC control panel. Clearing alarms in this manner also resets the "down" condition of the CRT terminal and permits the previously attempted operation to be re-tried.

A single corrective action is not always appropriate for all faults. In such cases, two or more indicators may be lit, suggesting possible actions to be taken and possibly indicating the nature of the fault. In these cases, the least severe actions should be taken first, followed by increasingly severe actions such as restart or reload. A list of operator actions arranged from least severe to most severe follows:

1. Clear alarms, then re-try current operation.
2. Restart the PMC by setting the PMC control panel MODE SELECTION switch to the OFF LINE position, then to the STANDBY or ON LINE position.
3. Reload the PMC (Refer to System Loading and Start up Instructions). If both restart and reload are indicated, the restart should be attempted first.
4. An illegal interrupt can occur spuriously during power fail/auto restart. If it persists, replace master station hardware and re-start.

In certain cases, the problem causing the alarm may not be severe enough to interfere with critical PMC control functions. In such cases, discretion should be used before applying the more severe corrective actions such as restart or reload.

In any case, if the indicated operator action(s) fails to clear the malfunction, maintenance personnel should be notified.

The following table applies to the SYSTEM FUNCTIONING lights on the PMC control panel. (See Figure 2-3).

SYSTEM FUNCTIONING Light	MEANING	OPERATOR ACTION
RUNNING light off	Processor stopped	Set MODE SELECTION switch OFF LINE, then back to ON LINE. If processor fails to come on, reload PMC.
TEMP. NORMAL light off	Cabinet temperature too high	Check for ventilation obstruction or faulty temperature sensor.
BAT. NORMAL light	Not applicable	Not applicable
CPU NORMAL light off	CPU power supply fault	Replace CPU power supply.
NO ALARM light out	Fault condition exists	Check CPU console lights for fault conditions described in the following table.

Indicators 8 through 15 on the CPU console display CSH error condition details. Those conditions serious enough to warrant taking the PMC off line also cause the CPU console indicators 0, 1 or 2 to be lit.

The following table applies to the panel lights on the CPU console. These panel lights are shown in Figure 2-4.

PANEL LIGHT	MEANING	OPERATOR ACTION
0	A restart may be necessary to clear the alarm condition.	Restart PMC by turning the PMC control panel MODE SELECTION switch to the OFF LINE position, then to STANDBY or ON LINE.
1	A reload may be necessary to clear the alarm condition.	Reload the PMC (Refer to System Loading and Startup Instructions). If both a restart and reload are indicated, the restart should be attempted first.
2	CSH reported fault. The PMC has been taken off line due to a serious CSH fault or malfunctioning I/O subsystem served by the CSH.	Note the error. Restart or reload as specified by indicators 0 and 1.
3	FSS DS-1000 fault.	Clear alarms, then re-try current operation.
4	IMM fault. This is probably a mini-cartridge tape problem.	Clear alarms, then re-try the current tape operation.
5	Printer fault. The PMC printer (if present) is not in a ready condition.	Correct printer problem, then clear alarms to resume operation.
6	The PMC control is not in a ready condition.	Correct control panel problem, then clear alarms to resume operation.
7	Spare	
8 - 14	Identify specific CSH reported fault condition. May or may not be serious enough to degrade process.	If accompanied by indicator 2, take action as for indicators 0 and 1. If not accompanied by indicator 2, see actions recommended below to interpret meaning.

PANEL LIGHT	MEANING	OPERATOR ACTION
0	A restart may be necessary to clear the alarm condition.	Restart PMC by turning the PMC control panel MODE SELECTION switch to the OFF LINE position, then to STANDBY or ON LINE.
1	A reload may be necessary to clear the alarm condition.	Reload the PMC (Refer to System Loading and Startup Instructions). If both a restart and reload are indicated, the restart should be attempted first.
2	CSH reported fault. The PMC has been taken off line due to a serious CSH fault or malfunctioning I/O subsystem served by the CSH.	Note the error. Restart or reload as specified by indicators 0 and 1.
3	FSS DS-1000 fault.	Clear alarms, then re-try current operation.
4	IMM fault. This is probably a mini-cartridge tape problem.	Clear alarms, then re-try the current tape operation.

5	Printer fault. The PMC printer (if present) is not in a ready condition.	Correct printer problem, then clear alarms to resume operation.
6	The PMC control is not in a ready condition.	Correct control panel problem, then clear alarms to resume operation.
7	Spare	
8 - 14	Identify specific CSH reported fault condition. May or may not be serious enough to degrade process.	If accompanied by indicator 2, take action as for indicators 0 and 1. If not accompanied by indicator 2, see actions recommended below to interpret meaning.

PANEL LIGHT(S)*	MEANING	OPERATOR ACTION
14 13		
X 0	Channel J Fault	Interpret indicator 12
X 0	Channel K Fault	Interpret indicator 12
X X	Channel L Fault	Interpret indicator 12
12		
0	Station error (I/O subsystem) malfunction	Interpret lighted indicators 8 through 11 as affected station number.
X	Channel error	Interpret lighted indicators 8 through 11 as error type explained below.
11 10 9 8		
0 0 0 0	CSH master fault	Correct problem, then take action per indicator 2
0 0 0 X	CSH error	If not accompanied by indicator 2, this is a spurious error. Frequent occurrences indicate pending serious problems. Perform maintenance on master/coax cables.
0 0 X 0	Excess CSH error rate.	Serious version of previous error. Will also be accompanied by a lighted indicator 2. Attempt restart and if this error is repeated, use the HTM to isolate the problem
0 0 X X	Channel off line.	Spurious error sometimes accompanying link error. See link error operator action (above).
0 X 0 0	Illegal interrupt.	Can occur spuriously during power fail/auto restart. Usually not serious. If it persists, replace master station hardware and restart
0 X 0 X	Illegal control word	Reload and restart.
0 X X X	Master communication error	Restart. If problem reoccurs, replace master station hardware.

*** 0 = light off, X = light on**

NORMAL SHUTDOWN PROCEDURE

1. Set PMC MODE SELECTION switch to OFF LINE position.
2. Unlock and open CPU front panel door.
3. Set POWER switch to OFF.
4. Close and lock CPU front panel door.
5. Set PS5 DC power supply switch CB1 to OFF.
6. Set AC POWER switch (on front of PMC control panel) to OFF.
7. Set BATTERY PACK EXT-INT-OFF switch (at rear of CPU) to OFF (This step is optional and results in loss of memory contents).

NORMAL STARTUP PROCEDURE (MEMORY CONTENTS RETAINED)

1. Set CPU POWER switch to OFF.
2. Set AC POWER switch to ON.
3. Set CPU POWER switch to ON.
4. Select PMC mode as desired.

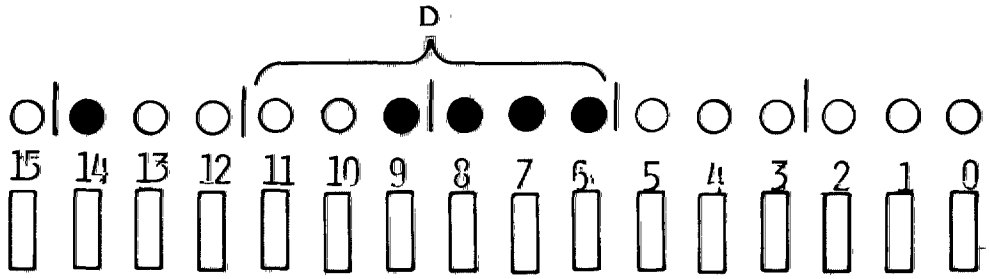
APPENDIX A
INSTRUCTIONS FOR USING IMM CASSETTE LOADER

On IMM Console:

1. Insert cassette tape into left-hand cassette drive.
2. Depress reset button twice at approximately a 1-second interval.

On CPU Console:

1. Depress HALT.
2. Select S/s register for display readout.
3. Depress CLEAR DISPLAY.
4. Set readout pattern 041700 in the display register as follows:



Where: D = device number of terminal (normally #17)

NOTE: Shaded circles represent illuminated display lights on CPU operator's console.

5. Depress STORE.
6. Depress PRESET.
7. Depress IBL.
8. Depress RUN.

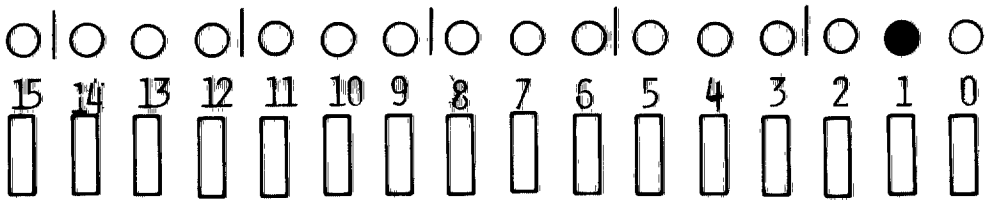
The cassette should begin loading as evidenced by the flashing green reject button. If the load is successful, the controller will halt (RUN light will go out) with octal 102077 appearing in the display register after approximately 2 minutes.

APPENDIX B
INSTRUCTIONS FOR INITIALIZING PMC

On CPU Operator's Console

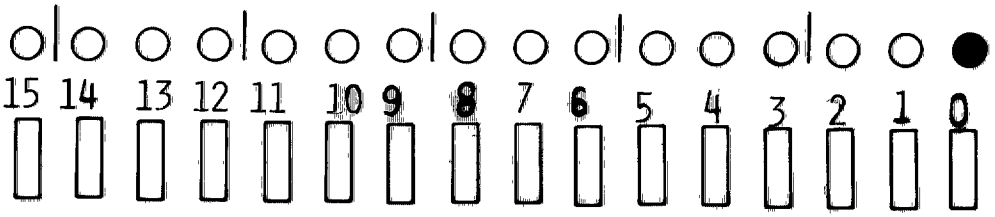
(See Figure 2-3).

1. Depress HALT.
2. Select P/f register for display.
3. Depress CLEAR DISPLAY.
4. Set the following readout pattern in the display register:



NOTE: Shaded circuit represents illuminated display light on CPU operator's console.

5. Depress STORE.
6. Select S/s register for display.
7. Set the following readout pattern in the display register:



8. Depress STORE.
9. Depress RUN.

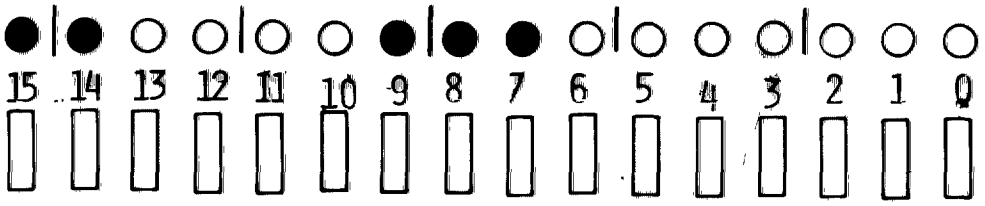
A successful load and initialization will be indicated by the appearance of "SET TIME", followed by a PROGRAM LIST on the IMM CRT screen.

APPENDIX C
INSTRUCTIONS FOR USING DATA LINK LOADER ROM

On CPU Console

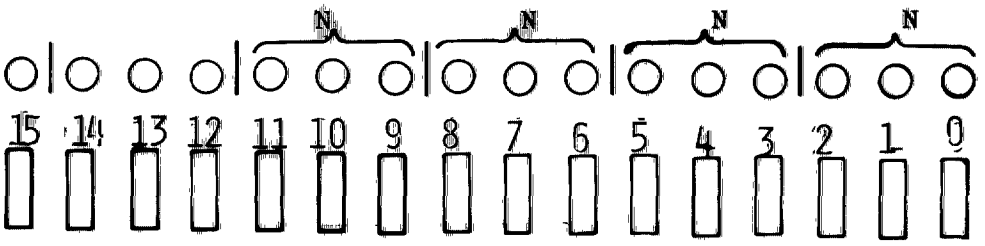
(Refer to Figure 2-3).

1. Depress HALT.
2. Select S/s register for display.
3. Depress CLEAR DISPLAY.
4. Set the following readout pattern in the display register:



NOTE:Shaded circles represent illuminated display lights on CPU console.

5. Depress STORE.
6. Depress PRESET.
7. Depress IBL.
8. Depress CLEAR DISPLAY.
9. Set the following readout pattern in the display register:



10. Depress STORE.

11. Depress RUN.

The down-line load will begin. The PMC will automatically initialize at the completion of the load. A successful load will be indicated by the following message appearing on the IMM CRT:

SET TIME

An automatic start-up command will be performed and the following message will then appear on the IMM CRT:

NNNN SYSTEM HAS BEEN BOOTED

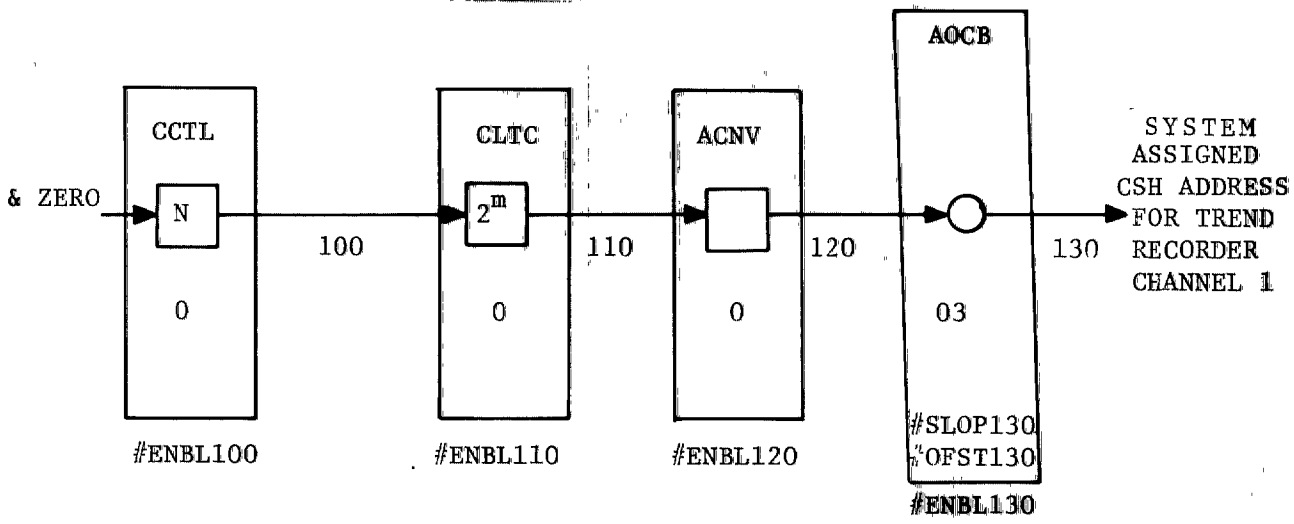
(Where: NNNN = Control system ID.)

APPENDIX D
SEGMENT USED FOR TREND COMMAND

CNST		140	CHANNEL	BLOCK #
#ENBL100	L	F	1	100-110-120-130-140
#ENBL110	L	F	2	200-210-220-230-240
#ENBL120	L	F	3	300-310-320-330-340
#ENBL130	L	F	4	400-410-420-430-440
#SLOP130	A	#1.0	5	500-510-520-530-540
#OFST130	A	#0.0	6	600-610-620-630-640
			7	700-710-720-730-740
			8	800-810-820-830-840
			9	900-910-920-930-940
			10	001-011-021-031-041
			11	101-111-121-131-141
			12	201-211-221-231-241

SEGMENT IDENTIFICATION

Segment Name: 999Z
Type: C
Rep. Rate: 10
SGABLE: &T
INABLE: &T
RNABLE: &T
OTABLE: &T



APPENDIX E
IMM ERROR MESSAGES

During routine operations on the IMM, an error message may appear on the CRT display. Error messages are of two major types:

General - consisting of the word ERROR followed by a 2-digit error code.

Relay Editing - consisting of the phrase RL ERROR followed by a 2-digit error code.

GENERAL ERROR MESSAGES

CODES	MEANING
(Codes 01-19 signify command monitor errors)	
02	INVALID NUMBER OF CHARACTERS IN COMMAND STRING
03	DS-1000 SEQUENCE ERROR IN SLAVE CALLS
04	BAD MEMORY ACCESS PARAMATER FROM FSS
05	DS-1000 SEQUENCE ERROR IN MASTER CALLS
06	DS-1000 HARDWARE ERROR
07	BUFFER OVERFLOW - FSS RETRIEVAL OF SIGNAL CURRENT VALUES
(Codes 20-29 signify command pre-processor (syntax) errors)	
20	INVALID COMMAND NAME
21	TOO FEW PARAMETERS IN COMMAND STRING
22	TOO MANY PARAMETERS IN COMMAND STRING
23	INVALID PARAMETER - IMBEDDED BLANKS
24	INVALID NUMBER OF CHARACTERS IN A PARAMETER
25	INVALID CHARACTER IN A PARAMETER
26	INVALID PARAMETER SEQUENCE
27	INVALID NUMBER - SYNTAX ERROR
28	NUMBER TOO LARGE
(Error codes 40-99 signify command handler errors)	
40	GIVEN SEGMENT ID NOT IN SSB
41	MODE CODE NOT 'E' OR 'D'
42	SEGMENT ALREADY IN REQUESTED MODE
43	ADDRESS OUT OF BOUNDS
44	INVALID DATA TYPE FOR THIS COMMAND
45	NUMBER OUT OF RANGE
46	NUMERIC OVERFLOW
47	GIVEN BLOCK NUMBER NOT IN SEGMENT
48	INVALID REQUEST - SEGMENT IS ENABLED
49	NO ROOM FOR NEW CONTROL BLOCK

50	BAD FORMAT IN FIRST WORD OF NEW CONTROL BLOCK
51	INVALID NUMBER OF WORDS IN NEW CONTROL BLOCK
52	CANNOT HAVE BLOCK NUMBER GREATER THAN OR EQUAL TO 1000
53	CANNOT DELETE 1ST BLOCK IN SEGMENT
54	BLOCK NUMBER LESS THAN OR EQUAL TO 1ST BLOCK IN SEGMENT
55	GIVEN POSITION NUMBER IN CONTROL BLOCK TOO LARGE
56	INVALID MODIFY, POSITION TO FORCE TABLE
57	CANNOT REPLACE/DELETE CONTROL BLOCK IN FORCE TABLE
58	MODIFY ENABLE KEY IN OFF POSITION
59	INVALID PACE OR OFFSET PARAMETER
60	DISPV, NUMBER OF LINES NOT BETWEEN 1 AND 1000
61	DISPLV, INVALID DATA TYPE
62	FORCE TABLE FULL
63	INVALID CONTROL BLOCK INSERT, BLOCK NUMBER ALREADY EXISTS.
64	FORCE 'USE' CODE NOT 'I' OR 'O'
65	INVALID EAM FLAG BITS IN SIGNAL ADRESS
66	BASED INDIRECT OFFSET FALLS OUTSIDE RCALL CONTROL BLOCK
67	INVALID 'LOP' FOR GIVEN 'USE' CODE
68	GIVEN SEG/BLK/POS NOT IN FORCE TBL
69	ENCOUNTERED 'PTN'OUTSIDE SEGMENT RANGE
70	INVALID CASSETTE DESIGNATOR
71	MAPPED MOVE ERROR (INTERNAL IMM PROGRAM ERROR)
72	INVALID PARAMETERS FOR SSGA CASSETTE
73	UNEXPECTED EOF ON CASSETTE
74	GIVEN SSGA RANGE NOT WITHIN SSGA IN MEMORY
75	INVALID PARAMETERS FOR PARTITION CASSETTE
76	INVALID CHANNEL NUMBER
77	DMOD BUFFER FULL (OVER 63 SEGMENTS IN SYSTEM)
78	ENCOUNTERED 2 IDENTICAL SEGMENT IDS IN DMOD COMMAND
79	INVALID TIME ENTERED FOR STIME COMMAND
80	INVALID DAY ENTERED FOR STIME COMMAND
81	INVALID DAY ENTERED FOR STIME COMMAND
82	INVALID YEAR ENTERED FOR STIME COMMAND
83	SPECIFIED CONTROL BLOCK IS NOT A RELAY LADDER CONTROL BLOCK
84	RELAY LADDER EDITING ABORTED BY OPERATOR
85	INVALID SIGNAL ADDRESS ON RELAY LADDER INSERT/MODIFY
86	INDEXING IS INVALID ON RELAY LADDERS
87	CONTROL BLOCK MACRO REQUIRES GREATER THAN 7 X 5 MATRIX
88	CHECKSUM ERROR IN RELAY LADDER TRANSMISSION
89	INVALID RELAY LADDER COMMAND
90	NO "RECALL" CONTROL BLOCK SPECIFIED FOR BASING
91	SEGMENT CURRENTLY LOCKED
92	COMMAND HANDLER NOT IMPLEMENTED

RELAY EDITING ERROR MESSAGES

The relay editing error messages which follow are of three types:

1. Invalid operator inputs

2. Transmission problems between the CPU and CRT due to one of the following:
   - bad cabling
   - loose connectors
   - terminal data - comm card malfunction
   - CPU hardware or software malfunction

3. CRT internal firmware or hardware malfunction. The CRT Self-Test should be executed. The error type is indicated in parentheses.

RL ERROR 2 - TRANSMIT TIME WARNING (Type 2)
This warning results from the terminal trying to transmit for 2.5 seconds and getting no response from the CPU. The terminal will try to transmit indefinitely.

RL ERROR 3 - DATA RECEPTION ERROR (Type 2)
Invalid data received from CPU.

RL ERROR 4 - RECEIVE TIME WARNING (Type 2)
This warning occurs when the terminal has waited for a response from the CPU for longer than 2.5 seconds.

RL ERROR 5 - INPUT BUFFER OVERFLOW (Type 3)
Data from the CPU has overflowed the terminal input buffer.

RL ERROR 6 - LRC CHECK ERROR (Type 2)
This indicates bad transmission of data from the CPU.

RL ERROR 7 - FIRST RELAY NOT IN CORNER (Type 1)
The first relay rung must start in the upper left corner.

RL ERROR 8 - OPEN CONTACT ERROR (Type 1)
One end of a contact is not connected to a rung.

RL ERROR 9 - SHORT CIRCUIT ERROR (Type 1)
There is a short around a contact.

RL ERROR 10 - REVERSE CURRENT ERROR (Type 1)
A contact has been inserted such that current can pass through it in either direction.

RL ERROR 11 - SORT ERROR #1 (Type 3)
Internal CRT software error.

RL ERROR 12 - SORT ERROR #2
Internal CRT software error.

RL ERROR 13 - SORT ERROR #3
Internal CRT software error.

RL ERROR 14 - ILLEGAL MESSAGE NUMBER (Type 3)
Internal CRT software error.

RL ERROR 15 - INVALID INSTRUCTION OP CODE (Type 2)
An invalid instruction op-code was encountered during the process of generating a relay ladder display.

RL ERROR 16 - RELAY LADDER REQ. 5 ROWS (Type 1)
The relay ladder to be displayed will not fit in the 5 row format of the CRT. This error is caused by the operator editing a relay ladder into a configuration which, when "normalized", will result in a relay ladder that requires more than 5 rows. Correction of this problem requires the operator to reconfigure the relay ladder into two KRLY control blocks.

RL ERROR 17 - RELAY LADDER REQN. 7 COLUMNS (Type 1)
The relay ladder, when normalized, requires more than 7 columns. Correction of this problem requires that the relay ladder be reconfigured into two KRLY blocks.

RL ERROR 18 - POPS NOT EQUAL PUSHES (Type 3)
This is an internal software error.

RL ERROR 19 - INVALID IMAGE DATA BASE OP CODE (Type 3)
Internal CRT software error.

APPENDIX F
MEMORY DISPLAY FUNCTION

The memory display function, consisting of four modes of operation, is controlled by a program called MEMZ which is a multi-purpose program designed for use by individuals having a detailed understanding of the internal PMC product software and operating system. MEMZ is used to observe internal operating conditions of the PMC. This function is normally dedicated to the display of PMC alarm conditions described in the **System Diagnostic Display** section. The MEMZ program is automatically activated during system start-up and remains continuously in operation.

This appendix describes the use of alternate modes of operation. It is important that each of the three modes described below transition to the PMC alarm mode after use.

1. ***Memory Display Mode***

   A. Base and Offset Registers

      MEMZ contains two special purpose address registers, MA and BA, which specify the address of the memory location to be displayed. BA is typically a base address to be used as the start of a block of memory. MA is an offset address which can be used as an offset into a block of memory. The effective address is calculated as the sum of BA + MA.

      These registers can be displayed by depressing and holding the top half of switch 15 (for MA) or the bottom half of switch 15 (for BA). Their contents can be modified while switch 15 is being held by entering the desired address in switches 14 through 0.

   B. Memory Display Mode

      This mode is entered when the top half of switch 15 is depressed and released. If the current effective address (BA + MA) is zero, the PMC alarm mode will be re-entered. Otherwise, the contents of the effective memory address will be displayed in the indicator lights.

   C. Stepping Through a Memory Block

      To display a series of memory locations in a block of memory, enter the starting address in BA and set MA equal to 0. The first memory location will now be displayed on the CPU console. To display the next location, momentarily depress and release the top half of switch 0. This will increment MA. To back up, depress and release the bottom half of switch 0.

2. PMC Freetime Calculation Mode

This mode calculates and displays the remaining CPU time available on the display indicators as a 4-digit BCD number representing 99.99% freetime (no load) to 00.00% freetime (fully loaded). To enter this mode, set MA and BA equal to 0, then depress and release the bottom half of switch 15. When first entered, the display will be cleared for 10 seconds. An initial estimate of the freetime will then be displayed. Evey 10 seconds thereafter, the calculation is repeated and filtered with a time constant of about 30 seconds and the new value is output. To exit this mode, change the state of indicator 15. After 10 seconds or less, the PMC alarm mode will be re-entered.

Measurements of the execution time of each CBL segment can be made by using the following technique:

1. Ensure that the segment to be measured is enabled using the IMM display mode (DMOD) function.
2. Enable the freetime calculation as descrived above. After 10 seconds, the initial estimate of remaining freetime will be displayed (i.e., 51.50%).
3. Wait for several cycles (30 seconds) to ensure that the final value has stabilized.
4. Disable the segment of interest using the IMM change mode (CMOD) function.
5. Re-enter the PMC alarm mode by changing the state of switch 15.
6. Re-enter the freetime calculation mode and repeat steps (2) and (3); note new freetime value (i.e., 55.63%).
7. Calculate the segment execution time (including CSH I/O) using the following formula:

$$ET = (DF-EF) \times RR \times .001$$

Where: ET = Execution time, milliseconds
DF = Disabled freetime, percent
EF = Enabled freetime, percent
RR = Segment repetition rate, milliseconds

E.G., $(55.63 - 51.5) \times 30 \times .001 = 1.239$ msec.

The execution time is similar to the calculated elapsed time. The calculated elapsed time can be calculated for each segment as follows:

$$ETC = 0.3 \times A. + 0.001(I + O) + N + 0.001$$

Where: ETC = Segment calculated execution time (milliseconds)

A = Number of CSH links (J,K or L) accessed for inputs plus the number of CSH links (J,K or L) accessed for outputs, e.g., if a segment does inputs from the J and K links and outputs to the K link, A equals 3.

I+O = Number of input addresses in the segment plus the number of output addresses.

N = Calculated block execution time (milliseconds).

APPENDIX G
ABBREVIATIONS

CB	Control block
CBL	Control block language
CIAD	CSH interactive and diagnostic program (test program)
COMCM	Common command monitor (program)
CSH	Control Signal Highway
EAM	Extended address modification
EOF	End of file
FSS	Field support system
FSSCM	Field support system command monitor (program)
IMM	Installation and maintenance module
LOP	Logic operator
PROM	Programmable ROM
PMC	Programmable master controller
PTN	Pointer to next control block
ROM	Read-only memory
SSGA	Sub-system global area

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